



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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ASSIGNMENT No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.



Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Input

```
EmployeeManager.java
1 package assignment8;
2 import java.util.Scanner;
3 class Employee
4 {
5     int empId;
6     String empName;
7     double salary;
8     Employee(int empId, String empName, double salary)
9     {
10         this.empId = empId;
11         this.empName = empName;
12         this.salary = salary;
13     }
14     void displayEmployee()
15     {
16         System.out.println("\nEmployee Details");
17         System.out.println("-----");
18         System.out.println("Employee ID : " + empId);
19         System.out.println("Employee Name : " + empName);
20         System.out.println("Salary : " + salary);
21     }
22 }
23 class Manager extends Employee
24 {
25     String department;
26     Manager(int empId, String empName, double salary, String department)
27     {
28         super(empId, empName, salary);
29         this.department = department;
30     }
31     void displayManager()
32     {
33     }
34 }
```

The screenshot shows the Eclipse IDE with the 'EmployeeManager.java' file open. The code defines two classes: 'Employee' and 'Manager'. 'Employee' has attributes 'empId', 'empName', and 'salary', and methods 'Employee(int empId, String empName, double salary)' and 'displayEmployee()'. 'Manager' extends 'Employee' and has an additional attribute 'department' and a constructor 'Manager(int empId, String empName, double salary, String department)' that calls 'super(empId, empName, salary)'. The 'displayManager()' method is also defined. The IDE interface includes a Package Explorer on the right showing the project structure, a Console at the bottom, and a Windows taskbar at the very bottom.



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```
EmployeeManager.java
class Manager {
    int empId;
    String empName;
    double salary;
    String department;

    void displayEmployee() {
        System.out.println("Employee ID : " + empId);
        System.out.println("Employee Name : " + empName);
        System.out.println("Salary : " + salary);
        System.out.println("Department : " + department);
    }
}

public class EmployeeManager {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Employee Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Salary: ");
        double salary = sc.nextDouble();
        sc.nextLine();

        System.out.print("Enter Department: ");
        String dept = sc.nextLine();

        Manager m = new Manager(id, name, salary, dept);
        System.out.println("\n===== Employee Information =====");
        m.displayEmployee();
        sc.close();
    }
}
```

Problems | Javadoc | Declaration | Console | Properties

terminated> EmployeeManager [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (01-Aug-2026, 14:08)

Enter Employee ID: 101

Enter Employee Name: Karan

Enter Salary: 1000000

Enter Department: HR

===== Employee Information =====

Employee Details

Employee ID : 101

Employee Name : Karan

Salary : 1000000.0

Department : HR

Output

```
Enter Employee ID: 101
Enter Employee Name: Karan
Enter Salary: 1000000
Enter Department: HR

===== Employee Information =====

Employee Details
-----
Employee ID : 101
Employee Name : Karan
Salary : 1000000.0
Department : HR
```



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2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Input

```
1 package assignment8;
2 import java.util.Scanner;
3 class Vehicle
4 {
5     int vehicleId;
6     String vehicleName;
7     String ownerName;
8     Vehicle(int vehicleId, String vehicleName, String ownerName)
9     {
10         this.vehicleId = vehicleId;
11         this.vehicleName = vehicleName;
12         this.ownerName = ownerName;
13     }
14     void displayVehicle()
15     {
16         System.out.println("\nVehicle Details");
17         System.out.println("-----");
18         System.out.println("Vehicle ID : " + vehicleId);
19         System.out.println("Vehicle Name : " + vehicleName);
20         System.out.println("Owner Name : " + ownerName);
21     }
22 }
23 class CarInsurance extends Vehicle
24 {
25     double insuranceAmount;
26     CarInsurance(int vehicleId, String vehicleName, String ownerName, double insuranceAmount)
27     {
28         super(vehicleId, vehicleName, ownerName);
29         this.insuranceAmount = insuranceAmount;
30     }
31     void displayInsurance()
32     {
33         super.displayVehicle();
34         System.out.println("Insurance Amount : " + insuranceAmount);
35     }
36 }
37 public class VehicleInsurance
38 {
39     public static void main(String[] args)
40     {
41         Scanner sc = new Scanner(System.in);
42         System.out.print("Enter Vehicle ID: ");
43         int id = sc.nextInt();
44         sc.nextLine();
45         System.out.print("Enter Vehicle Name: ");
46         String vehicleName = sc.nextLine();
47         System.out.print("Enter Owner Name: ");
48         String owner = sc.nextLine();
49         System.out.print("Enter Insurance Amount: ");
50         double amount = sc.nextDouble();
51         CarInsurance car = new CarInsurance(id, vehicleName, owner, amount);
52         System.out.println("\n==== Vehicle Insurance Details =====");
53         car.displayInsurance();
54         sc.close();
55     }
56 }
57
```

terminated> VehicleInsurance [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (01-Aug-2026, 2)

14:13 01-08-2026



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Output

```
Enter Vehicle ID: 6998
Enter Vehicle Name: Ertiga
Enter Owner Name: Karan
Enter Insurance Amount: 10000

===== Vehicle Insurance Details =====

Vehicle Details
-----
Vehicle ID : 6998
Vehicle Name : Ertiga
Owner Name : Karan
Insurance Amount : 10000.0|
```